

Title: Operating Systems Exam

Course Code: IT412

Instructor: Dr. Muhanna

Duration: 60

Q1. Which of the following is an example of a real-time operating system? (mcq) [1]

- a) Windows 10
- b) VxWorks
- c) macOS
- d) Ubuntu

Correct Answer: B

Q2. What is the main purpose of an operating system? (mcq) [1]

- a) To provide internet access
- b) To manage hardware and software resources
- c) To compile programs
- d) To store user files

Correct Answer: B

Q3. Which scheduling algorithm gives priority to the shortest job? (mcq) [1]

- a) FCFS
- b) Round Robin
- c) SJF
- d) Priority Scheduling

Correct Answer: C

Q4. Which of the following causes a deadlock? (mcq) [1]

- a) Too much memory
- b) Circular wait among processes
- c) Fast CPU speed
- d) Large disk storage

Correct Answer: B

Q5. Which memory management technique divides memory into fixed-size blocks? (mcq) [1]

- a) Segmentation
- b) Paging
- c) Swapping
- d) Compaction

Correct Answer: B

Q6. A process in the 'waiting' state is currently using the CPU. (true_false) [0.5]

- a) True

b) False

Correct Answer: B

Q7. Virtual memory allows a computer to run programs larger than physical RAM.

(true_false) [0.5]

a) True

b) False

Correct Answer: A

Q8. The kernel is the core component of an operating system. (true_false) [0.5]

a) True

b) False

Correct Answer: A

Q9. Round Robin scheduling uses a fixed time quantum for each process. (true_false)

[0.5]

a) True

b) False

Correct Answer: A

Q10. A semaphore can only have binary values (0 or 1). (true_false) [0.5]

a) True

b) False

Correct Answer: B